Insuroo: AI-Based Virtual Insurance Agent for Rural Areas

A Mini Project Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

Computer Science and Engineering

by

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CERTIFICATE

This is to certify that the report entitled **Insuroo: AI-Based Virtual Insurance Agent for Rural Areas** submitted by **ABHIRAM A P** (SCT23CS006), **AKSHAY M NAIR** (SCT23CS014), **ASHA LAKSHMI** (SCT23CS021) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering is a bonafide record of the mini project work carried out by them under our guidance and supervision. This report has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the mini project report **Insuroo: AI-Based Virtual Insurance Agent for Rural Areas**, submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by us under the supervision of **Dr. Soniya B**, Project Guide.

This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also invoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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Abstract

This project presents an AI-powered virtual insurance agent aimed at improving insurance awareness and accessibility in rural India, where challenges such as low literacy, complex policy terms, and limited advisory services hinder adoption. The primary objective is to provide a user-friendly digital platform that simplifies insurance concepts and guides users in selecting suitable policies.

The system employs a Retrieval Augmented Generation (RAG) approach, which combines a pre-trained language model with verified insurance documents and government scheme data to generate accurate and context-aware responses. A conversational interface enables interaction through text or voice, ensuring ease of use for diverse users. Additionally, an AI-powered recommendation module leverages user profile information and retrieval-augmented generation (RAG) techniques to suggest personalized insurance categories, enhancing decision-making.

The solution integrates a lightweight frontend with a FastAPI-based backend and an AI layer comprising document retrieval and an explainable recommendation engine. The outcome is a scalable and practical system that improves understanding, builds trust, and promotes insurance adoption in underserved communities. Overall, the project demonstrates how AI can be effectively leveraged for social impact by bridging the information gap in rural insurance access.

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Chapter 1

Introduction

Insurance plays a crucial role in providing financial protection against uncertainties related to health, agriculture, life, and property. Despite its importance, insurance penetration in rural India remains significantly low. This gap is primarily due to factors such as lack of awareness, complex policy structures, limited access to professional advisors, and language as well as literacy barriers. As a result, a large segment of the rural population remains vulnerable to financial risks.

With the rapid advancement of Artificial Intelligence (AI), there is an opportunity to address these challenges through intelligent and accessible digital solutions. This project proposes an **AI-Based Virtual Insurance Agent** designed specifically to improve insurance accessibility and understanding among rural communities. The system aims to simplify complex insurance concepts and provide personalized guidance through an easy-to-use conversational interface that supports both **text and voice-based interaction**.

The proposed solution integrates modern AI techniques such as Natural Language Processing (NLP), speech processing, and Retrieval-Augmented Generation (RAG) to deliver accurate and context-aware responses. By grounding the system's outputs in verified insurance documents and government schemes, it ensures reliability and trustworthiness. Additionally, an AI-driven recommendation module leverages user-specific data and retrieval-augmented generation techniques to provide personalized insurance recommendations tailored to individual needs.

The inclusion of a voice-based assistant enables users to interact with the system using natural speech, making it highly accessible for individuals with low literacy levels. This enhances user engagement and ensures inclusivity, particularly in rural settings where typing-based interfaces may not be effective.

The system is implemented using a lightweight frontend interface and a FastAPI-based backend, combined with a pre-trained language model and an explainable recommendation engine. This architecture ensures scalability, efficiency, and user-friendliness, making it suitable for deployment in real-world rural environments.

By combining accessibility, personalization, accuracy, and voice-enabled interaction, this project aims to bridge the information gap in rural insurance adoption and demonstrate the effective use of AI-driven solutions in public welfare domains. Ultimately, it contributes toward empowering rural populations with better financial awareness and decision-making capabilities.

Chapter 2

Literature Review

This chapter presents a review of existing research works related to the application of Artificial Intelligence in the insurance domain, particularly focusing on accessibility, recommendation systems, and rural outreach. The analysis of these works helps in identifying research gaps and motivates the proposed system.

2.1 Leveraging artificial intelligence to improve health insurance access and address disparities in rural Africa

Methodology: This study presents a conceptual analysis of Artificial Intelligence applications in rural health insurance systems. It examines the role of machine learning, automation, and predictive analytics in improving insurance services. The research also reviews key challenges associated with insurance accessibility in rural communities, including awareness, infrastructure, and service delivery.

Key Results: The study demonstrates that AI can significantly improve insurance accessibility in rural areas. Automation enhances the efficiency of claims processing and policy management, while AI-based models contribute to effective fraud detection and risk assessment. Overall, the approach helps in reducing healthcare disparities.

Limitations: The study is primarily theoretical and lacks practical implementation details. It focuses mainly on African health insurance systems, which may limit its applicability to other regions. Additionally, it assumes the availability of strong digital infrastructure, which is often lacking in rural areas.

Relevance to Our Work: This work supports the use of AI-based solutions for improving insurance accessibility in rural regions. It highlights the importance of intelligent assistance systems, which aligns with our proposed AI-based virtual insurance agent designed for rural users.

2.2 Artificial intelligence service agents: a silver lining in rural India

Methodology: This study presents a case analysis of Kotak Life's AI chatbot avatar KAYA (AISA) used for rural insurance services. The research is based on a sample of 707 rural customers from southern India, selected using multi-stage cluster sampling. Data analysis was performed using tools such as SPSS and SmartPLS, employing techniques like percentage analysis, multiple regression, and Structural Equation Modeling (SEM). The study evaluates key variables including efficiency, security, availability, enjoyment, and customer interaction, and their impact on economic and market performance.

Key Results: The findings indicate that efficiency and security are the most significant factors influencing system performance. Other important dimensions include availability, customer interaction, and user experience. The chatbot system showed a considerable improvement in claim-processing quality and overall convenience for rural users.

Limitations: The study is limited to a single insurer (Kotak Life) and a specific chatbot system (KAYA). It focuses only on rural customers in southern India, which may limit generalization. Additionally, the analysis is based on cross-sectional and self-reported data.

Relevance to Our Work: This study demonstrates that rural users can effectively interact with AI-based insurance agents. It emphasizes the importance of optimizing system performance in terms of speed, security, and 24/7 availability. It also provides a useful evaluation framework that can be adopted in our project for performance assessment.

2.3 Application of Artificial Intelligence and Machine Learning in the Indian Insurance Industry

Methodology: This study analyzes the adoption of Artificial Intelligence (AI) and Machine Learning (ML) in the Indian insurance sector. It is based on secondary data collected from IRDAI reports covering the period from 2018 to 2023. The research examines various AI applications including claims automation, fraud detection, and customer support systems.

Key Results: The findings reveal that AI significantly improves operational efficiency and enhances customer service in the insurance industry. Chatbots and automation technologies reduce processing time for claims and policy-related services. Additionally, predictive analytics contributes to better risk assessment and enables the development of personalized insurance plans.

Limitations: The study focuses primarily on industry-level AI adoption and does not address rural accessibility challenges. It does not propose a user-centric digital platform for improving insurance awareness. Furthermore, there is limited discussion on populations with low digital literacy.

Relevance to Our Work: This study supports the implementation of AI-based systems in insurance services. Our project builds upon these findings by developing an AI-powered virtual insurance agent tailored for rural communities. It emphasizes user education, policy recommendation, and improved accessibility.

2.4 Machine Learning Recommendation System for Health Insurance Decision Making in Nigeria

Methodology: This study develops a machine learning-based recommendation system for health insurance decision-making. It utilizes algorithms such as K-Nearest Neighbors (KNN) and cosine similarity to generate recommendations. The system considers user-specific data including age, income, and health condition to suggest appropriate insurance policies.

Key Results: The system effectively recommends suitable insurance policies tailored to individual user profiles. It enables users to compare different insurance plans and improves overall decision-making efficiency in selecting insurance coverage.

Limitations: The system requires structured input data, which may limit usability for non-technical users. Additionally, it lacks natural language processing capabilities and does not support voice or conversational interaction.

Relevance to Our Work: This study supports the use of AI-based recommendation systems in insurance services. Our project enhances this approach by integrating an AI chatbot with a Retrieval-Augmented Generation (RAG) system, enabling more interactive, user-friendly, and context-aware insurance recommendations.

2.5 Artificial Intelligence in Insurance Sector

Methodology: This study explores various AI technologies applied in the insurance sector, including recommendation engines, Natural Language Processing (NLP)-based chatbots, virtual assistants, context-aware systems, and predictive APIs for fraud detection and risk assessment. It analyzes real-world implementations such as AXA using Google TensorFlow for motor loss prediction, Fukoku Mutual Life Insurance using IBM Watson for claims processing, and Transamerica using H2O.ai for cross-selling strategies.

Key Results: The findings show that AI significantly enhances operational performance across insurance processes. For example, Fukoku achieved a 30% increase in claims processing productivity, while AXA reached 78% accuracy in motor loss prediction. AI also supports personalized sales through cross-selling strategies, as demonstrated by Transamerica. Additional benefits include fraud reduction, real-time pricing, and improved customer service through chatbots. The study also highlights rapid market growth in AI adoption, with spending increasing from $4.8 billion in 2016 to $47 billion in 2020.

Limitations: The study identifies several challenges, including data privacy and ownership concerns, ethical issues, and high implementation costs. Integration with legacy systems is also a significant barrier for many insurance companies.

Relevance to Our Work: This study reinforces the importance of integrating ML and NLP technologies in insurance systems. Our project applies these concepts to develop prototypes for PMJJBY claims processing and fraud detection in rural Kerala. The conceptual model can be adapted for building MVP chatbot systems, enhancing accessibility and aligning with ESG-focused insurance initiatives in India.

Chapter 3

Requirement Analysis

This chapter outlines the functional and non-functional requirements of the AI-Based Virtual Insurance Agent. The system is designed to address the challenges of low insurance awareness, complex policy structures, and limited accessibility in rural and semi-urban regions.

3.1 Problem Analysis

Despite the availability of numerous insurance schemes in India, a large portion of the population remains uninsured or underinsured. The key challenges identified are:

- Lack of awareness about available insurance policies and government schemes
- Complexity in understanding policy terms, benefits, and eligibility criteria
- Language and literacy barriers, especially in rural areas
- Limited access to financial advisors and insurance agents
- Difficulty in comparing multiple policies and making informed decisions

The proposed system aims to overcome these challenges by providing an intelligent, accessible, and user-friendly virtual assistant.

3.2 System Objectives

The primary objectives of the system are:

- To provide personalized insurance recommendations based on user profiles
- To simplify complex insurance policies into understandable insights
- To support both voice and text-based interaction for improved accessibility
- To ensure accurate and reliable responses using RAG-based retrieval
- To enable real-time interaction with minimal latency

3.3 Functional Requirements

The functional requirements define the core features and capabilities of the system.

3.3.1 User Interaction

- The system shall allow users to interact using both text and voice input
- The system shall provide a conversational chat interface
- The system shall support multilingual interaction

3.3.2 User Data Collection

- The system shall collect user details such as age, income, and occupation
- The system shall dynamically adapt survey questions based on previous inputs
- The system shall store user profile data for personalization

3.3.3 Query Processing

- The system shall process natural language queries using an LLM
- The system shall convert voice input into text using STT
- The system shall generate responses in both text and speech formats

3.3.4 RAG-Based Information Retrieval

- The system shall retrieve relevant insurance documents from a vector database
- The system shall use semantic search for accurate context retrieval
- The system shall generate responses grounded in retrieved data

3.3.5 Recommendation Engine

- The system shall analyze user profiles to recommend suitable insurance policies
- The system shall rank policies based on relevance and eligibility
- The system shall provide explanations for each recommendation

3.3.6 Output Presentation

- The system shall display recommendations in a structured format
- The system shall highlight key benefits, eligibility, and coverage details
- The system shall provide voice output for accessibility

3.4 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

3.4.1 Performance

- The system should respond to user queries within 2–5 seconds
- The system should handle multiple concurrent users efficiently

3.4.2 Scalability

- The system should support scaling to thousands of users
- The architecture should allow easy addition of new modules or services

3.4.3 Usability

- The interface should be simple and intuitive for non-technical users
- The system should support voice interaction for low-literacy users

3.4.4 Reliability

- The system should provide accurate and consistent responses
- The system should minimize hallucinations using RAG architecture

3.4.5 Security and Privacy

- User data should be securely stored and transmitted
- The system should ensure privacy of personal and financial information

3.4.6 Maintainability

- The system should be modular and easy to update
- Logs and monitoring should be implemented for debugging

3.5 System Constraints

- Dependence on external APIs such as LLM, STT, and TTS services
- Internet connectivity required for real-time processing
- Limited computational resources on user devices

3.6 Assumptions

- Users have access to a smartphone with internet connectivity
- Users can provide basic personal information for recommendations
- Insurance data sources are accurate and regularly updated

3.7 Summary

The requirement analysis highlights the need for an intelligent, accessible, and scalable system that simplifies insurance decision-making. By addressing both functional and non-functional aspects, the proposed system ensures a robust solution capable of real-world deployment.

Chapter 4

System Development

This chapter presents the design, architecture, and implementation of the AI-Based Virtual Insurance Agent. The system is developed to bridge the accessibility gap in rural insurance literacy by delivering accurate, personalized, and easy-to-understand insurance guidance through a multimodal conversational interface supporting both text and voice interactions.

The system simplifies complex insurance policies and government schemes into actionable insights, enabling users to make informed decisions regardless of their technical or financial knowledge. It is designed with a strong focus on inclusivity, scalability, and real-time responsiveness, making it suitable for deployment in resource-constrained environments.

4.1 System Architecture

The system follows a modular four-tier architecture designed for scalability, flexibility, and low-latency performance. Each layer operates independently, allowing efficient updates, debugging, and maintenance without affecting the overall system.

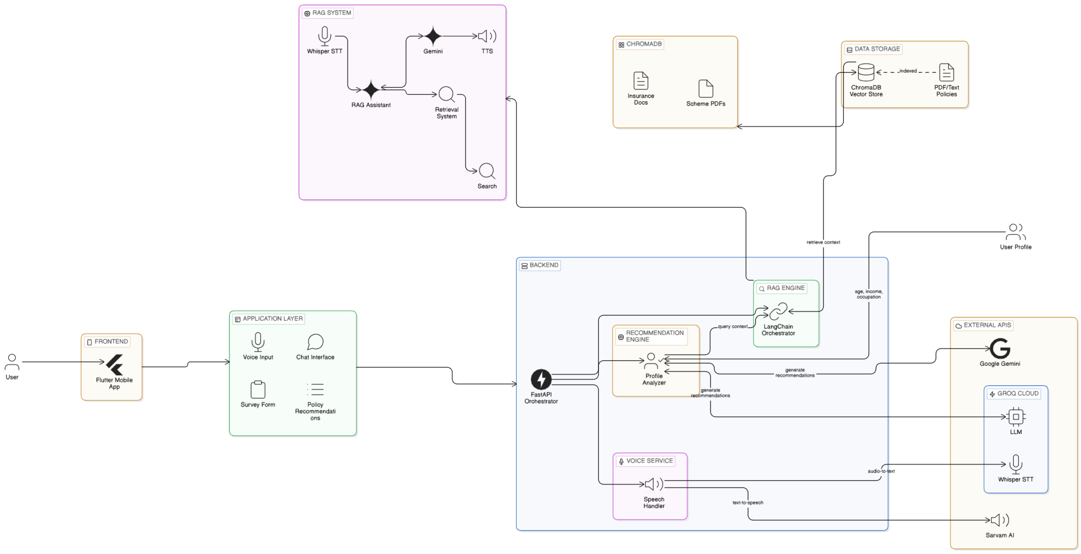


Figure 4.1: System Architecture of AI-Based Virtual Insurance Agent

- **User Interface Layer:** This layer acts as the primary interaction point between the user and the system. Developed using Flutter, it supports cross-platform deployment (Android and iOS). The interface is optimized for low-literacy users through intuitive UI elements, voice support, and minimal text complexity. It includes a multimodal chat interface, dynamic survey forms for collecting user data, and a dashboard for displaying structured policy recommendations in a user-friendly manner.

- **Backend Orchestration Layer:** Implemented using FastAPI, this layer serves as the central control unit of the system. It handles API routing, asynchronous request processing, session management, and communication between different modules. It ensures efficient handling of concurrent user requests and maintains contextual continuity across conversations.

- **AI Processing Layer:** This layer forms the intelligence core of the system and is responsible for understanding user queries and generating responses. It includes:
  - Large Language Model: Google Gemini 3.0 Flash for fast and context-aware reasoning
  - RAG Framework: LangChain for orchestrating retrieval and generation pipelines

  - Vector Database: ChromaDB for storing semantic embeddings
  - Embedding Model: Sentence-Transformers for converting text into vector representations

  This layer enables semantic understanding, contextual reasoning, and retrieval of accurate information from domain-specific knowledge sources.

- **Voice Processing Layer:** This layer enables speech-based interaction, improving accessibility for users with limited literacy. It consists of:
  - Speech-to-Text (STT): Whisper-large-v3 via Groq for real-time transcription
  - Text-to-Speech (TTS): SarvamAI for natural and regional language audio output

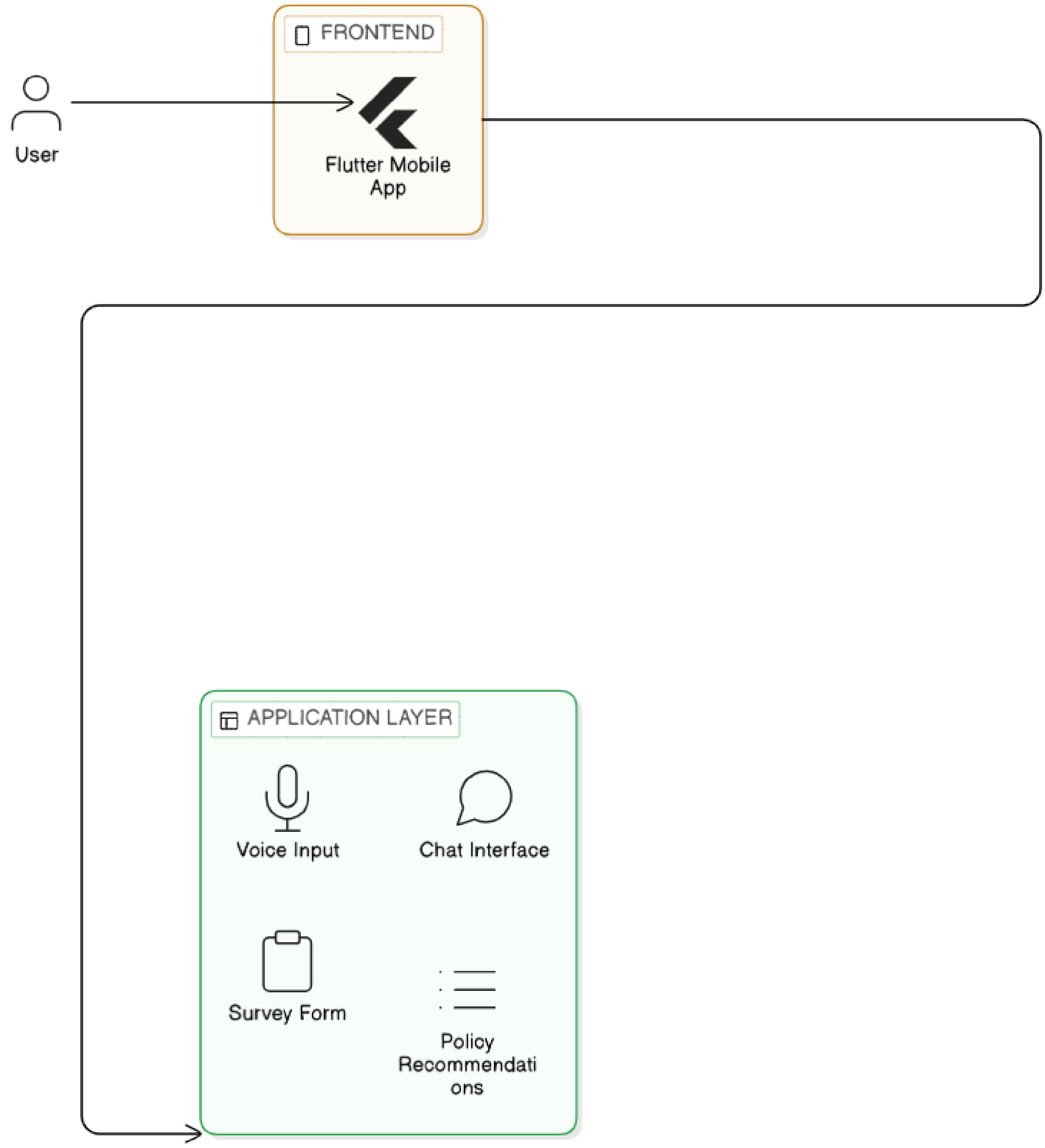
  The integration of voice technologies allows seamless bidirectional communication between the user and the system.

4.2 User Interface and Application Layer

The application layer follows a mobile-first design approach to ensure usability across diverse user groups, especially in rural settings. The design emphasizes simplicity, clarity, and accessibility.

- **Multimodal Chat Interface:** The chat interface allows users to interact using either text or voice. It supports conversational AI capabilities, enabling natural language queries. The system maintains conversational context to provide coherent responses across multiple interactions.
- **Dynamic Survey Form:** The survey module collects essential user information such as age, income level, occupation, family size, and risk preferences. The form dynamically adapts based on previous inputs to reduce user effort and improve data accuracy.

- **Policy Recommendation Dashboard:** This module presents recommended insurance policies in a structured format, including benefits, eligibility criteria, premium details, and coverage. It also highlights key differences between policies to aid decision-making.



4.3 Backend Orchestration

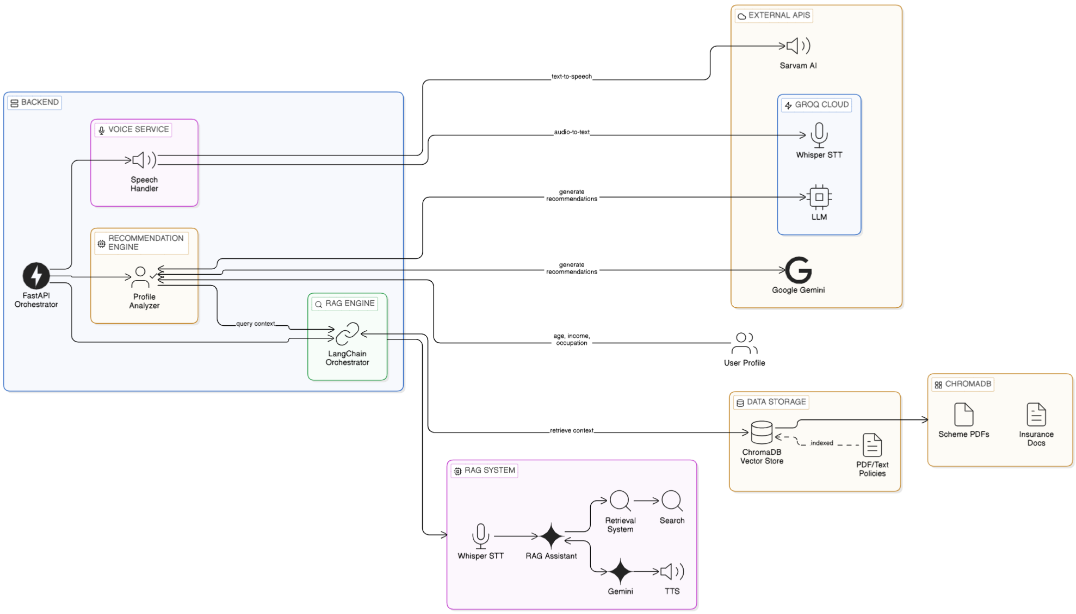
The backend functions as an asynchronous controller that manages system operations and communication between modules.

- **Request Routing:** Incoming user requests are analyzed and routed to the appropriate modules such as the RAG engine, voice processing service, or recommendation engine.

- **Context Management:** The backend maintains user session data, including conversation history and profile information. This enables the system to generate context-aware responses and ensures continuity in multi-turn conversations.

- **Service Integration:** The backend integrates with external APIs such as LLM providers, STT engines, and TTS services. It handles API calls efficiently and manages latency through asynchronous processing.

- **Error Handling and Logging:** Robust error handling mechanisms are implemented to ensure system reliability. Logs are maintained for debugging and performance monitoring.



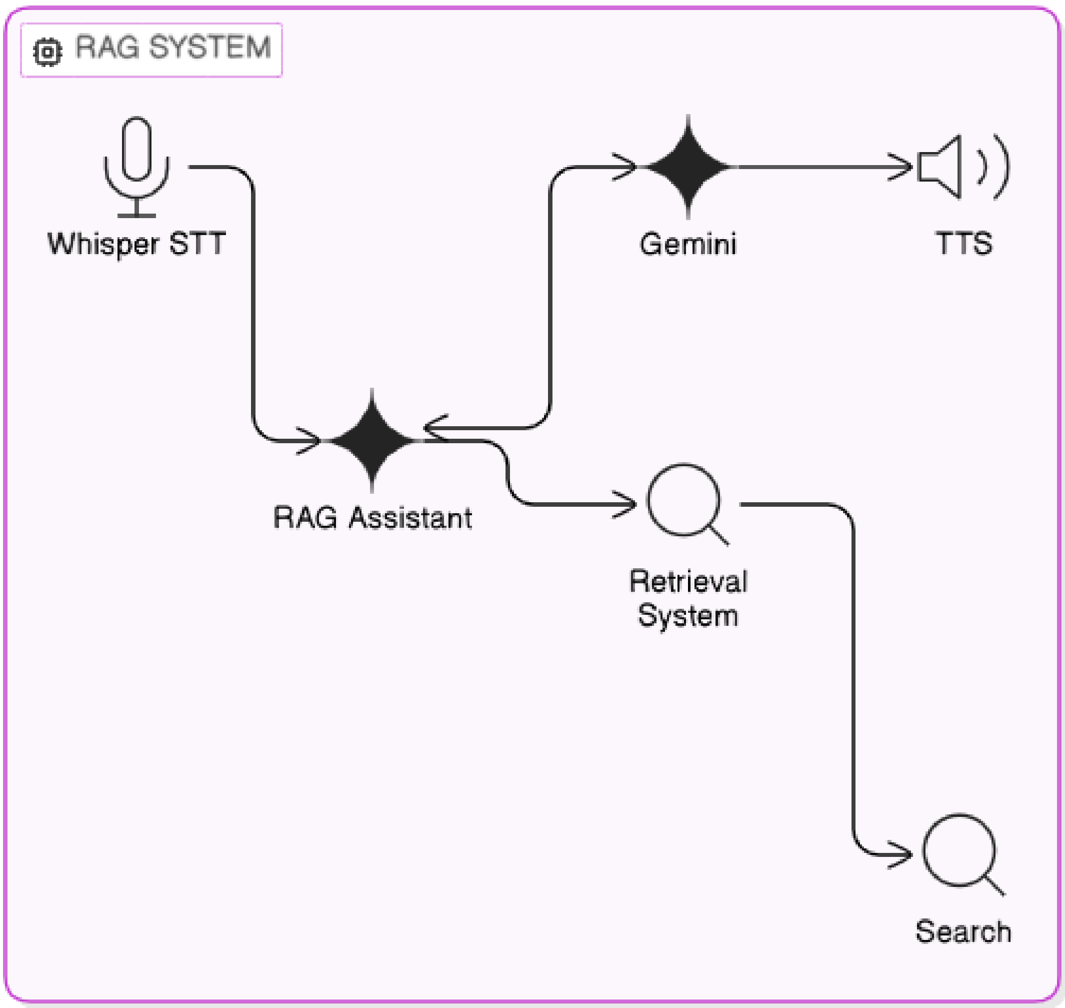
4.4 Retrieval-Augmented Generation (RAG)

The system employs a Retrieval-Augmented Generation (RAG) framework to ensure accuracy, reliability, and domain-specific correctness of responses.

- Insurance documents and government policy data are preprocessed through text cleaning, segmentation, and chunking

- Each document chunk is converted into embeddings using Sentence-Transformers
- Embeddings are stored in ChromaDB for efficient similarity search
- User queries are transformed into embeddings and matched using semantic search techniques
- Relevant document chunks are retrieved and passed as contextual input to the LLM
- The LLM generates responses grounded in retrieved knowledge, reducing hallucinations

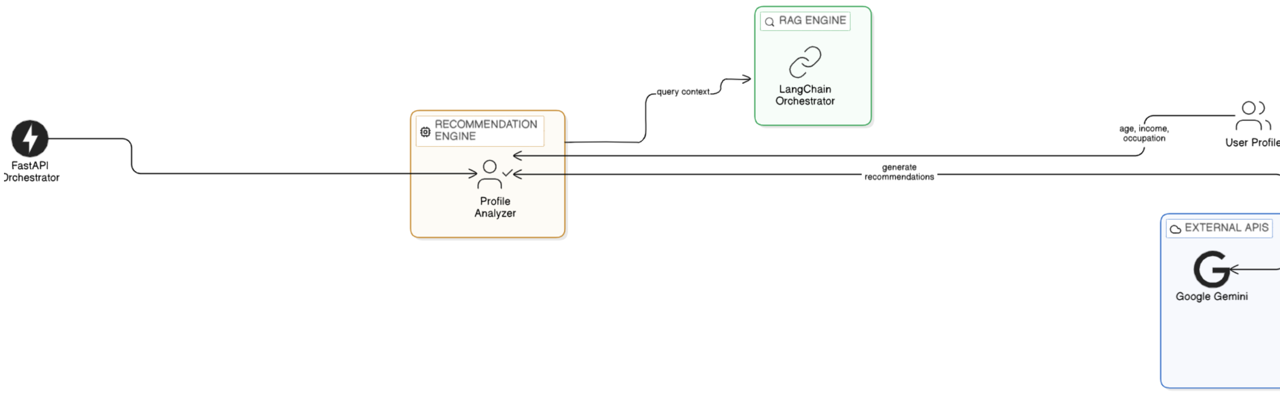
This hybrid retrieval-generation approach significantly enhances factual accuracy and ensures that responses are aligned with verified policy information.



4.5 Personalized Recommendation Engine

The recommendation engine is designed to provide highly relevant and personalized insurance suggestions tailored to individual user profiles.

- **User Profiling:** The system collects demographic and socio-economic data such as age, income, occupation, and family structure.
- **Eligibility Matching:** The engine compares user attributes against eligibility criteria of various insurance schemes and filters out unsuitable options.
- **Ranking and Scoring:** Policies are ranked based on relevance, affordability, and coverage benefits.
- **Explainable Recommendations:** The system generates human-readable explanations to justify why a particular policy is recommended, improving user trust and transparency.



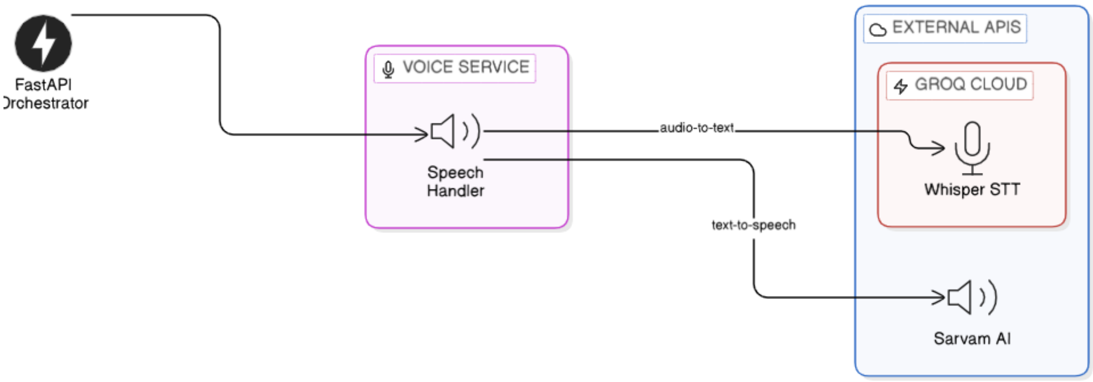
4.6 Voice Service Module

The voice module enhances accessibility by enabling natural speech interaction.

- User speech input is captured and transmitted as audio data
- Audio is converted into text using Whisper STT via Groq
- The transcribed text is processed by the AI system
- The generated response is converted back into speech using SarvamAI TTS

- Audio output is delivered to the user in a natural and understandable format

This module significantly reduces barriers for users who may not be comfortable with reading or typing.



4.7 Data Storage and Knowledge Base

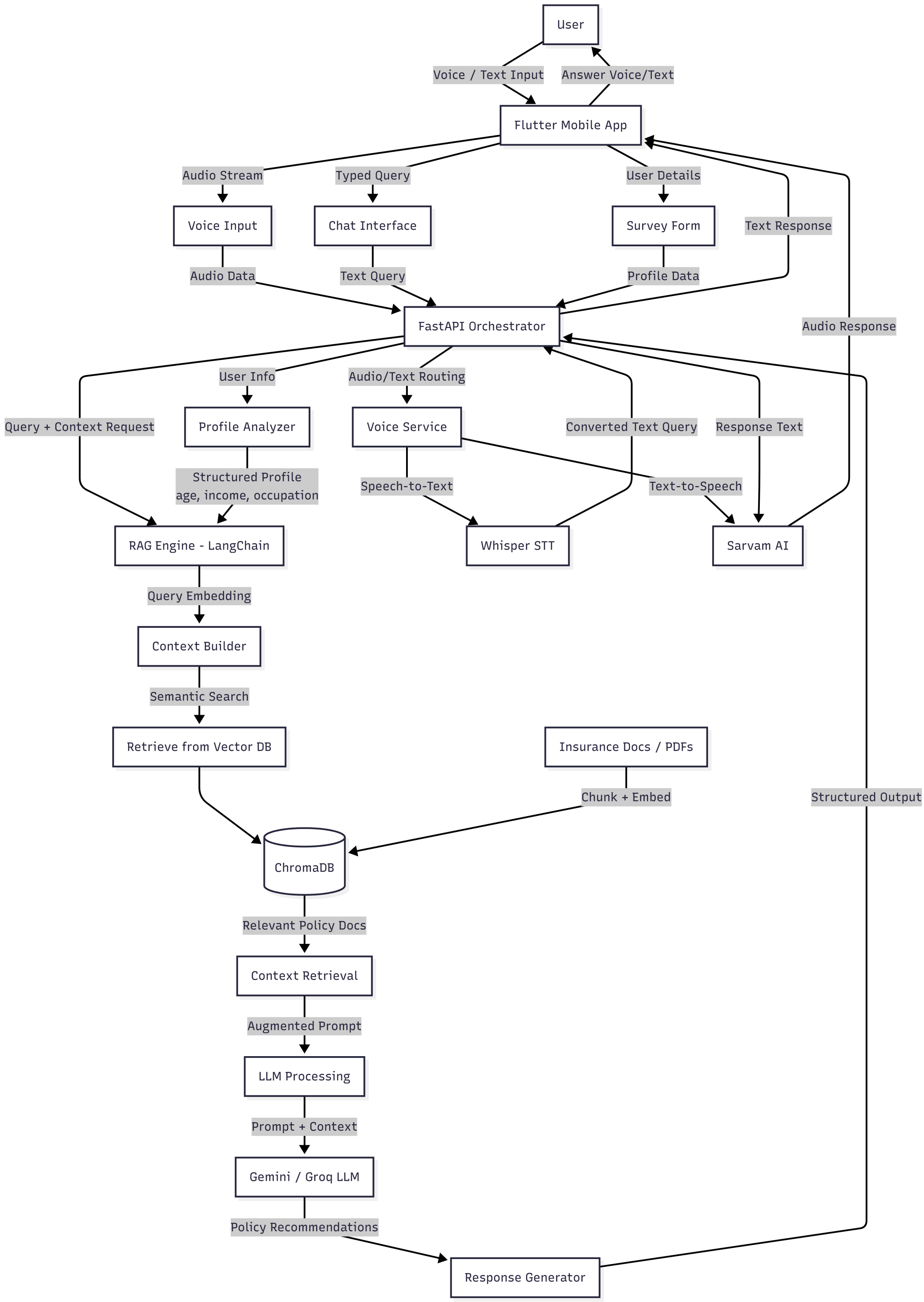
The system relies on a structured and scalable data layer for accurate and efficient information retrieval.

- **Vector Store:** ChromaDB is used to store high-dimensional embeddings of insurance documents, enabling fast semantic search.
- **Document Repository:** Contains verified and curated insurance policies, government schemes, and guidelines. These documents serve as the authoritative knowledge base.
- **Indexing Pipeline:** Documents are periodically updated, reprocessed, and indexed to ensure that the system remains up-to-date with the latest policy information.

4.8 System Workflow

The overall workflow of the system integrates multiple components into a seamless pipeline:

1. The user provides input through text or voice via the mobile application
2. Voice input is converted into text using the STT module
3. The backend processes the request and retrieves relevant user profile information
4. The query is forwarded to the RAG engine for contextual understanding
5. Relevant documents are retrieved from the vector database using semantic search
6. The LLM generates a context-aware and personalized response
7. The response is returned in text format and optionally converted into speech
8. The final output is displayed to the user through the application interface



4.9 Summary

The system integrates multiple advanced AI technologies into a unified platform that delivers accessible, accurate, and personalized insurance assistance. The combination of RAG-based retrieval, voice-enabled interaction, and intelligent recommendation mechanisms ensures high usability and reliability.

The modular architecture, scalability, and real-time processing capabilities make the system suitable for deployment in rural and underserved communities, thereby contributing to increased insurance awareness and financial inclusion.

Chapter 5

Technology Stack and System Design

This chapter outlines the technologies and tools used in the development of the AI-Based Virtual Insurance Agent. The selected stack ensures scalability, efficiency, and seamless integration of AI, backend services, and user interaction modules.

5.1 Frontend Technologies

- **Flutter:** A cross-platform UI toolkit used to build a responsive and mobile-first application. It enables the development of a single codebase for multiple platforms with a consistent user experience.
- **Dart:** The programming language used with Flutter for building interactive user interfaces and handling client-side logic.

5.2 Backend Technologies

- **Python:** The primary programming language used for backend development due to its strong support for AI and web frameworks.
- **FastAPI:** A high-performance web framework used to build APIs for handling requests, managing sessions, and orchestrating communication between system components.

5.3 AI and Machine Learning Technologies

- **Google Gemini 2.5 Flash:** A Large Language Model (LLM) used for generating intelligent, context-aware responses.
- **LangChain:** A framework used to implement the Retrieval-Augmented Generation (RAG) pipeline by combining document retrieval with LLM-based response generation.
- **Sentence-Transformers:** Used to generate vector embeddings for semantic search and similarity matching.

5.4 Data Storage and Retrieval

- **ChromaDB:** A vector database used to store embeddings of insurance documents and enable fast semantic retrieval.
- **Document Repository:** A structured storage system containing insurance policies, government schemes, and related documents used as the knowledge base.

5.5 Voice Processing Technologies

- **Whisper-large-v3 (via Groq):** Used for Speech-to-Text (STT) conversion, enabling accurate transcription of user voice input.
- **SarvamAI:** Used for Text-to-Speech (TTS) conversion, generating natural and clear spoken responses.

5.6 Integration and Deployment

- **API Integration:** External APIs are used for LLM inference, speech processing, and other AI services.

- **Modular Architecture:** The system is designed with loosely coupled components, allowing independent scaling and updates.

5.7 API Design and Routes

The backend system is implemented as a RESTful API using FastAPI. The API layer exposes endpoints for query processing, recommendation generation, and voice interaction. These endpoints enable seamless communication between the frontend and backend services.

Table 5.1: API Endpoints and Descriptions

Method	**Endpoint**	**Description**
GET	/health	Checks if the API server is running
POST	/query/ask	Processes user queries using the RAG pipeline
POST	/query/recommend	Generates personalized insurance recommendations
POST	/voice/transcribe	Converts speech input into text
POST	/voice/speak	Converts text into speech

5.8 Data Models

The system defines structured data models for handling requests and responses between the frontend and backend.

- **QuestionRequest**
  - query: string
- **UserProfile**
  - age: integer
  - income: float
  - occupation: string
- **AnswerResponse**
  - answer: string

- sources: list of documents

- **RecommendationResponse**
  - policies: list of recommended schemes
  - explanation: string

5.9 Summary

The chosen technology stack combines modern development frameworks with advanced AI tools to deliver a scalable, efficient, and intelligent system. Each component plays a critical role in ensuring smooth operation, accurate information retrieval, and user-friendly interaction.

Chapter 6

Results and Discussion

This chapter presents the evaluation, testing results, and analysis of the AI-Based Virtual Insurance Agent. The system was assessed based on performance, accuracy, robustness, and usability to determine its effectiveness in improving insurance accessibility, particularly for rural users.

6.1 Testing and Evaluation Metrics

The system was evaluated using multiple performance metrics across its core modules, including the chatbot, voice interface, RAG system, and recommendation engine.

6.1.1 Chatbot and Voice Module

- **Transcription Accuracy (Word Error Rate):** Measures the accuracy of speech-to-text conversion. The system achieved approximately 85–92% accuracy for Hindi and mixed-language inputs.
- **Intent Recognition Accuracy:** Evaluates how effectively user queries are mapped to appropriate backend actions. The system achieved approximately 90% accuracy for structured queries.
- **Response Latency:** The average response time ranged between 1.5 to 3 seconds for text queries and 3 to 5 seconds for voice-based interactions.

- **Graceful Error Handling:** The system successfully handled API failures and missing configurations without crashing, ensuring robust user experience.

6.1.2 RAG System Performance

- **Retrieval Context Relevance (Precision):** Achieved approximately 88% precision in retrieving contextually relevant insurance information from the vector database.

- **Answer Faithfulness (Hallucination Rate):** The use of Retrieval-Augmented Generation significantly reduced hallucinations by grounding responses in retrieved documents.

- **Answer Relevance:** Generated concise, accurate, and context-aware responses aligned with user queries.

- **Cold-Start vs Hot-Start Latency:** Subsequent queries (hot-start) showed up to 40% faster response times compared to initial system initialization.

6.1.3 Recommendation Engine Performance

- **Constraint Matching Accuracy:** Successfully matched user profiles with policy eligibility criteria such as age and income.

- **Personalization Quality:** Provided tailored recommendations with clear justifications based on user inputs.

- **Edge Case Robustness:** Demonstrated stable performance even with incomplete or out-of-range input data.

6.2 System Implementation Results

The developed system successfully integrates conversational AI, voice interaction, and retrieval-based knowledge systems into a unified platform. Users can interact using both text and voice inputs to obtain relevant insurance information.

Experimental evaluation demonstrates that the system provides accurate, context-aware, and personalized responses using the RAG approach. The system maintains low latency while ensuring high-quality output.

6.3 Chatbot Performance

The chatbot effectively understands user queries and generates meaningful responses.

- Accurate interpretation of insurance-related queries
- Reduced ambiguity through document-grounded responses
- Ability to handle multi-turn conversational queries

6.4 Voice Assistant Performance

The voice module enhances accessibility by enabling natural speech interaction.

- Reliable speech-to-text conversion for query processing
- Natural and intuitive interaction experience
- Improved usability for users with low literacy levels

6.5 Recommendation System Results

The recommendation engine provides personalized insurance suggestions based on user inputs.

- Accurate policy recommendations aligned with user profiles
- Improved decision-making through structured insights
- Faster identification of relevant insurance schemes

6.6 System Output Screenshots

This section presents screenshots of the working application, demonstrating key functionalities including voice interaction, conversational AI, and personalized insurance recommendations.

6.6.1 Voice Interaction

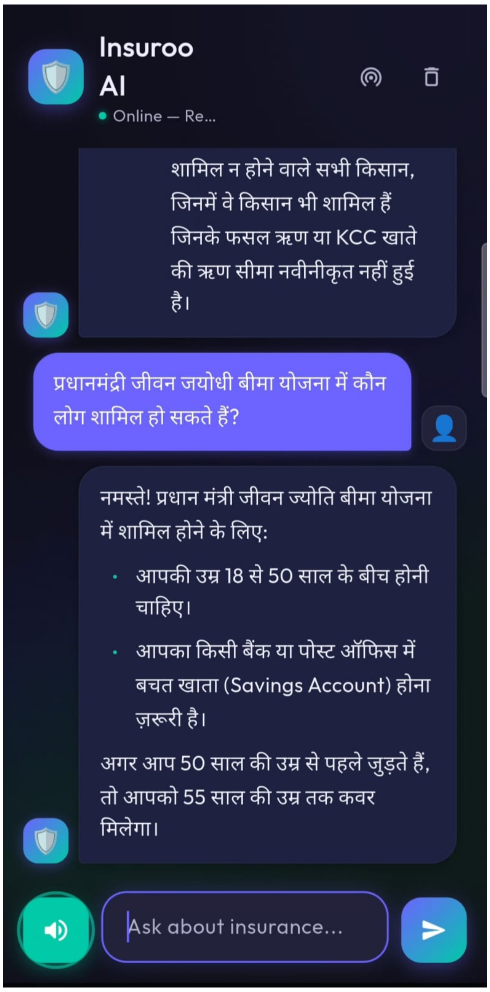


Figure 6.1: Voice-Based Interaction: User input through speech and system response

6.6.2 Text-Based Chat Interface

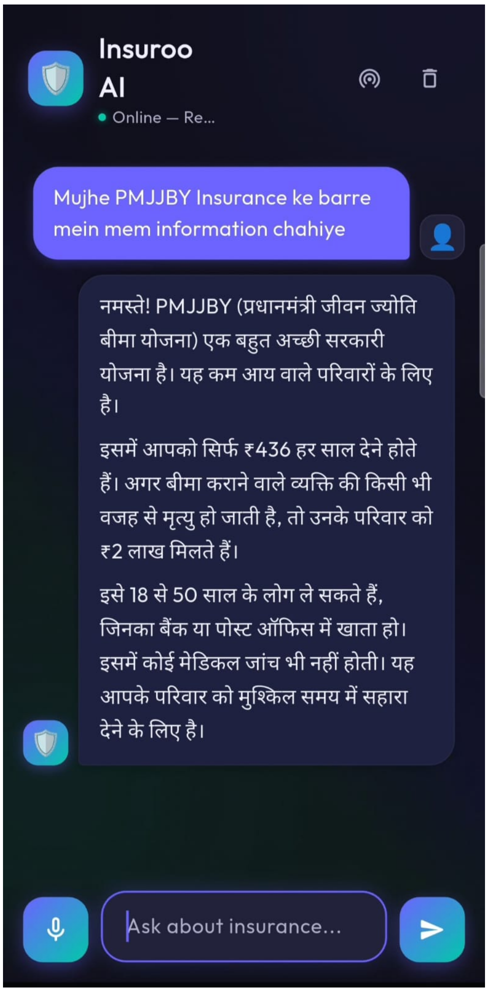


Figure 6.2: Chat Interface demonstrating conversational query handling

6.6.3 Recommendation Engine Outputs

The following screenshots illustrate the personalized insurance recommendations generated based on different user profiles.

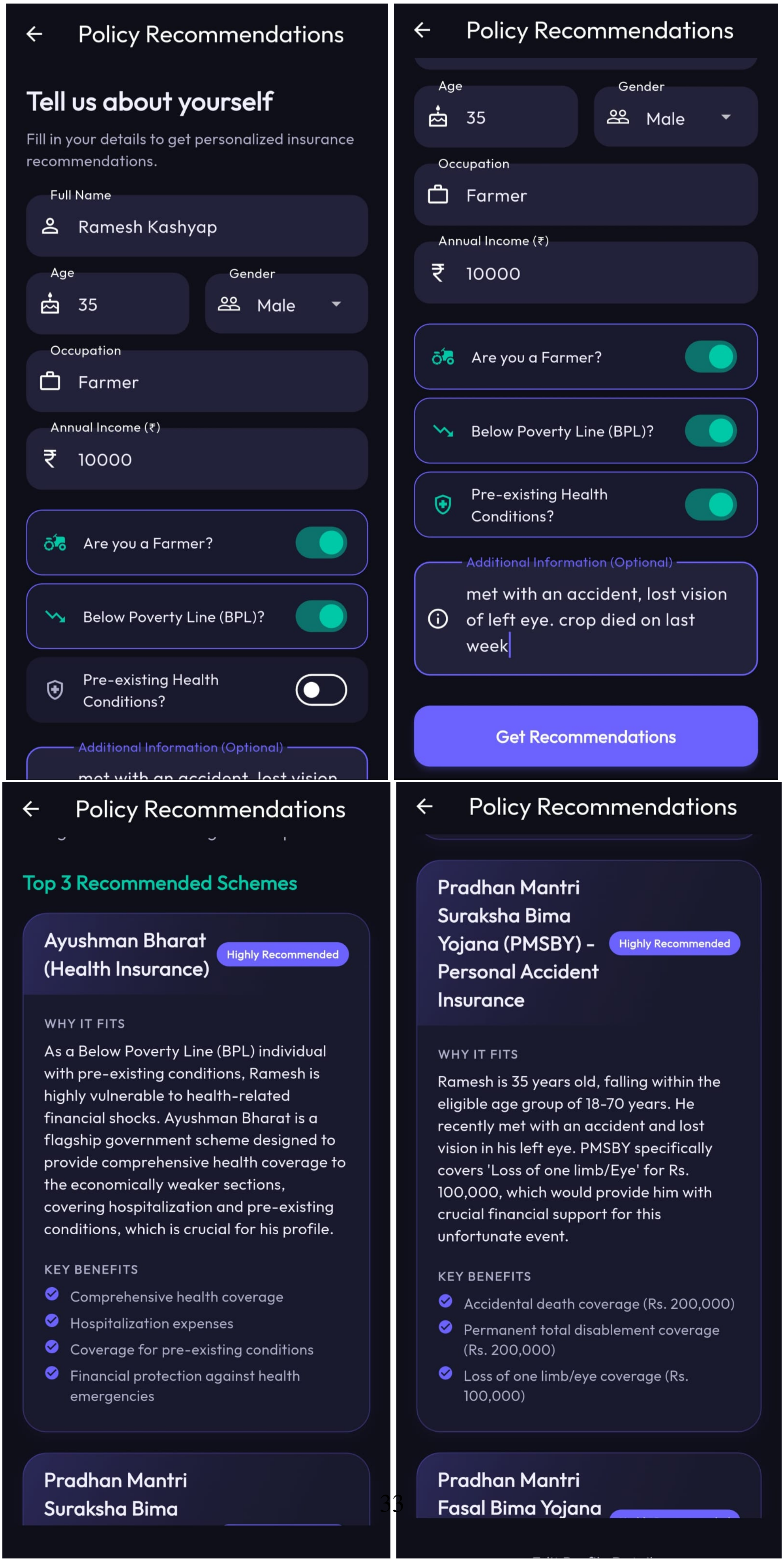


Figure 6.3: Recommendation Engine outputs

6.7 System Usability

The system interface is designed to be simple and accessible.

- User-friendly interface for both text and voice interaction
- Minimal learning curve for new users
- Suitable for deployment in rural environments

6.8 Discussion

The results demonstrate that the proposed system effectively addresses key challenges in insurance accessibility. The integration of Retrieval-Augmented Generation improves factual accuracy and reduces misinformation compared to traditional chatbot systems.

The voice-enabled interface significantly enhances usability, particularly for users with limited literacy, making the system inclusive and practical for real-world deployment.

However, certain limitations were identified:

- Dependence on stable internet connectivity for API-based services
- Variability in speech recognition accuracy for diverse regional accents
- Limited multilingual support in the current implementation

Future improvements can focus on offline capabilities, enhanced regional language support, and fine-tuning domain-specific AI models to further improve performance.

Chapter 7

Conclusion

This project presented the design and development of an **AI-Based Virtual Insurance Agent** aimed at improving insurance accessibility for rural populations. The system addresses key challenges such as low awareness, complex policy structures, language barriers, and limited access to professional guidance in underserved regions.

By integrating Artificial Intelligence techniques such as Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and machine learning-based recommendation systems, the proposed solution delivers accurate, simplified, and personalized insurance-related information. The system features a conversational interface that supports both **text and voice-based interaction**, enabling users to communicate naturally and making it especially beneficial for individuals with low literacy levels.

The inclusion of a voice-based assistant enhances accessibility by allowing users to interact with the system through speech, thereby reducing dependency on reading and typing skills. This makes the system more inclusive and user-friendly for rural communities. Additionally, the recommendation module analyzes user profiles to suggest suitable insurance options, improving decision-making and user satisfaction.

The implementation of a lightweight frontend and a FastAPI-based backend ensures scalability, efficiency, and ease of deployment. The system demonstrates how AI-driven digital assistants can effectively bridge the gap between insurance providers and rural users, promoting financial inclusion and improving service delivery.

However, certain limitations exist, including dependency on internet connectivity, challenges in handling multiple regional languages and dialects in voice interaction, and the need for extensive real-world testing. Future enhancements can include advanced multilingual voice support, offline functionality, integration with government insurance schemes, and continuous model improvement based on user feedback.

In conclusion, the proposed AI-based virtual insurance agent, enhanced with voice interaction capabilities, provides a practical and scalable solution for improving insurance awareness and accessibility in rural areas, showcasing the transformative potential of Artificial Intelligence in public welfare applications.

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